

GCC Sustainability Game Changer Hackathon

Explore NYC Open Data • Analyze a challenge • Recommend a solution

Student orientation | R or Python pathway

Before we get started:

- NYC Open Data: https://opendata.cityofnewyork.us
- NYC API guide: https://www.nyc.gov/opendata/get-started/user-guides/open-data-apis
- RStudio / Posit: https://posit.co/downloads
- Posit Cloud: https://posit.cloud
- Python: https://www.python.org

Our challenge

Use publicly available NYC OpenData to investigate a sustainability problem.

Produce an evidence-based recommendation with a chart and a short pitch.

Success is measured by insight and community impact.



What you will learn



1. Identify a meaningful NYC sustainability question
2. Locate and evaluate a dataset in NYC Open Data
3. Run a starter script in R or Python
4. Interpret a summary and chart
5. Communicate a feasible recommendation

What is data Analysis?

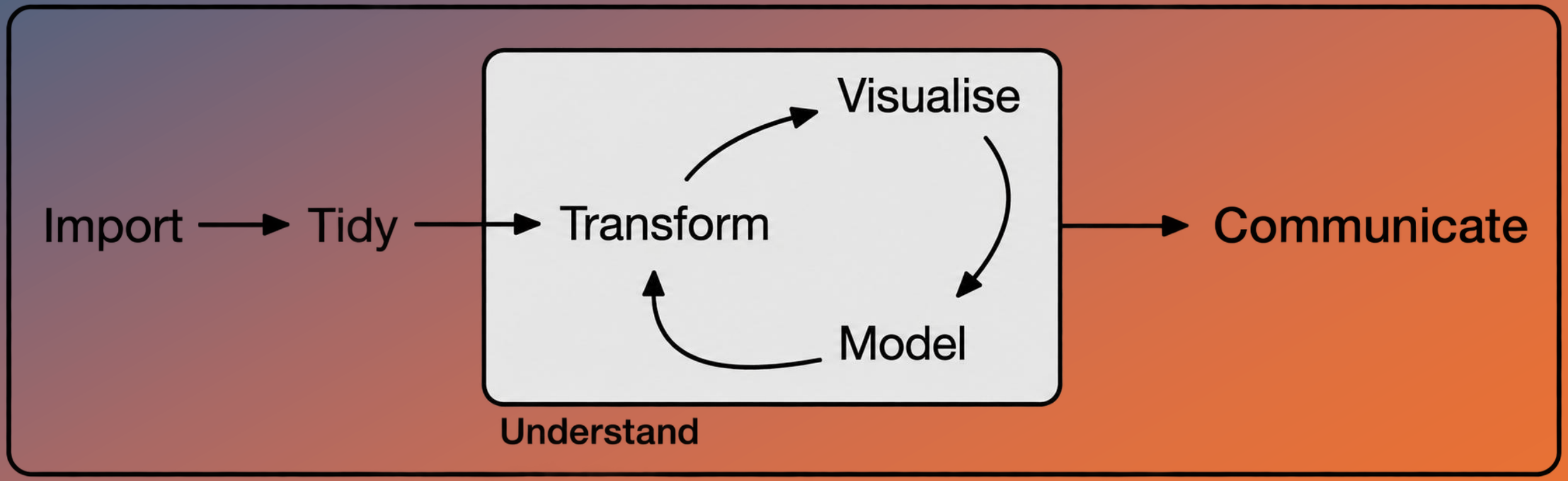
DATA
Collection of Items

→

PROCESS
Analysis
Calculation

→

INFORMATION
Understanding
Solutions



Data Analysis Workflow

Sustainability questions to explore



- Environment: How are trees distributed across neighborhoods?
- Waste: Where might recycling outreach be needed?
- Transportation: Where are mobility or safety concerns concentrated?
- Buildings & energy: What patterns might inform energy-efficiency outreach?
- Water & resilience: Where should further investigation be prioritized?

Start with the question - not the code



QUESTION → DATASET → FIELDS → CLEANING → SUMMARY → CHART → RECOMMENDATION

Example: “How do the number of trees vary by borough?”

The question drives the solution. R and Python are just tools, like pencil and paper, that help us get there.

Find your dataset

1. Visit opendata.cityofnewyork.us
2. Select Data and search your topic
3. Open a dataset and read its description, update date, and fields
4. Confirm that the dataset contains rows and columns
5. Find its unique ID (for example: uvpi-gqnh)
6. Note the URL for your citation

A helpful first dataset

- 2015 Street Tree Census – Tree Data
- NYC Open Data dataset ID: uvpi-gqnh
- Example fields: boroname, spc_common, health, tree_id
- This is a historical snapshot, NOT a count of all current NYC trees.
- Dataset: data.cityofnewyork.us/d/uvpi-gqnh

What is an API?



An API (application program interface) is a way for two software applications to communicate and exchange information without a person having to manually transfer the data.

Dataset ID + API address → CSV response → DataFrame

What is an API?



Example URL:

https://data.cityofnewyork.us/resource/uvpi-gqnh.csv?$limit=5000

The row limit keeps the first practice run manageable.

One workflow. Two languages.

Both starter scripts have the same settings and the same output:

1. Dataset ID and dataset name
2. Row limit
3. Category column (grouping)
4. Optional numeric value (average)
5. Top N categories

Output: summary.csv and chart.png

R and Python: shared idea

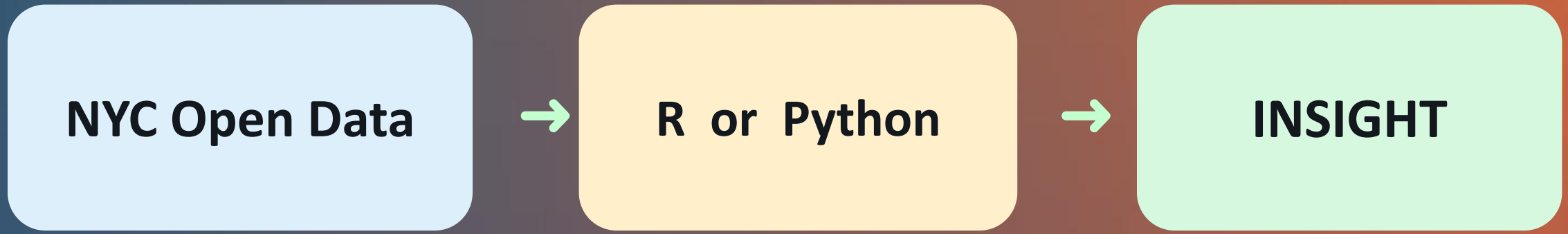
Both starter scripts have the same settings and the same output:

- R uses a data frame; Python uses a pandas DataFrame.
- Both store data as rows and columns.
- Both can group, count, calculate averages, and create charts.
- Choose one pathway for your team; do not switch mid-project.

Data → Evidence → Insight → Action

What Are R and Python?

Both are programming languages that help us collect, organize, analyze, and visualize data.



Same goal: turn city data into evidence for a sustainability recommendation.

What Is an Object?

In both languages, an object holds a value or dataset, with type and other information that helps the computer work with it.

DATA + **TYPE / METADATA**

A variable is the name we use to refer to the object.

Example: the number 10 is different from the text "10".

Variables: Give Objects a Name

Same idea, slightly different assignment syntax.

R

```
x <- 10
```

Python

```
x = 10
```

The name x refers to a value of 10. The language knows it is numeric.

Working with a Dataset: Data Frames

Both languages can store tabular data with rows (records) and columns (attributes).

R data frame

df <- read_csv(url)

pandas DataFrame

df = pd.read_csv(url)

In both scripts, df is the object containing the downloaded dataset.

Packages in R • Libraries in Python

Additional tools give each language specialized data-analysis capabilities.

R

readr — import CSV
dplyr — tidy and summarize
ggplot2 — create charts

Python

pandas — import and analyze
matplotlib — create charts
pathlib — manage file paths

Students choose one language; the analytical steps stay the same.

One Dataset. Two Ways to Download.

Example: NYC 2015 Street Tree Census • dataset ID uvpi-gqnh

R

```
df <- readr::read_csv(
download_url)
```

Python

```
df = pd.read_csv(
download_url)
```

The NYC Open Data API returns CSV records that become a data frame.

Explore, Select, and Tidy

Start by examining the available columns. Choose the ones that answer your question.

R

```
names(df)
select(df, boroname, health)
```

Python

```
df.columns.tolist()
df[["boroname", "health"]]
```

Check missing values, spelling, duplicates, and whether the records are comparable.

Analyze, Visualize, Communicate

The final product is not simply code or a chart; it is an evidence-based recommendation.

ANALYZE
Group and summarize

→

VISUALIZE
Create a readable chart

→

COMMUNICATE
Findings + limitation + action

Explain what the chart shows, what it cannot prove, and what NYC could do next.

Run the starter code:

C:\WINDOWS\system32

```
 Directory of P:\scratch_pad\GCC_Hackathon\20261026_Hackathon\hackathon_code

10/24/2026  07:19 PM    <DIR>          .
10/24/2026  05:19 PM    <DIR>          ..
10/24/2026  05:51 PM            11,294 GCC_Sustainability_Hackathon_Python_Completed_Example.py
10/24/2026  07:14 PM             4,804 GCC_Sustainability_Hackathon_Python_Starter_Code.py
10/24/2026  05:17 PM            15,330 GCC_Sustainability_Hackathon_R_Completed_Example.R
10/24/2026  07:21 PM             6,593 GCC_Sustainability_Hackathon_R_Starter_code.R
10/24/2026  07:19 PM    <DIR>          hackathon_results_python
               4 File(s)         38,021 bytes
               3 Dir(s)  4,394,346,057,728 bytes free

P:\scratch_pad\GCC_Hackathon\20261026_Hackathon\hackathon_code>py GCC_Sustainability_Hackathon_Python_Starter_Code.py
STEP 1: Output folder: \\gccfiler\users$\RNedd\scratch_pad\GCC_Hackathon\20261026_Hackathon\hackathon_code\hackathon_results_python

STEP 2: COLLECT
Downloading: https://data.cityofnewyork.us/resource/uvpi-gqnh.csv?$limit=1000
Downloaded shape: (1000, 45)
 tree_id  block_id              created_at  tree_dbh  stump_diam curb_loc status health                        spc_latin  spc_common steward guards sidewalk        user_typ
e problems root_stone root_grate root_other trunk_wire trnk_light trnk_other brch_light brch_shoe brch_other          address  zipcode     zip_city  cb_num  borocode boronam
e  cncldist  st_assem  st_senate  nta          nta_name  boro_ct     state  latitude  longitude        x_sp        y_sp  council_district  census_tract       bin          bbl
  180683    348711 2015-08-27T00:00:00.000         3           0   OnCurb  Alive   Fair                      Acer rubrum   red maple     NaN    NaN NoDamage TreesCount Staf
f      NaN         No         No         No         No         No         No         No         No         No 108-005 70 AVENUE    11375 Forest Hills     406         4   Queen
s       29         28         16 QN17      Forest Hills  4073900 New York 40.723092 -73.844215 1027431.148 202756.7687             29.0         739.0 4052307.0 4022210001.0
  200540    315986 2015-09-03T00:00:00.000        21           0   OnCurb  Alive   Fair                 Quercus palustris     pin oak     NaN    NaN   Damage TreesCount Staf
f   Stones        Yes         No         No         No         No         No         No         No         No  147-074 7 AVENUE    11357   Whitestone     407         4   Queen
s       19         27         11 QN49       Whitestone  4097300 New York 40.794111 -73.818679 1034455.701 228644.8374             19.0         973.0 4101931.0 4044750045.0
  204026    218365 2015-09-05T00:00:00.000         3           0   OnCurb  Alive   Good Gleditsia triacanthos var. inermis honeylocust    1or2     NaN   Damage        Voluntee
r      NaN         No         No         No         No         No         No         No         No 390 MORGAN AVENUE    11211    Brooklyn     301         3 Brookly
n       34         50         18 BK90 East Williamsburg  3044900 New York 40.717581 -73.936608 1001822.831 200716.8913             34.0         449.0 3338310.0 3028870001.0

STEP 3: EXPLORE
  1. tree_id
  2. block_id
  3. created_at
  4. tree_dbh
  5. stump_diam
  6. curb_loc
  7. status
  8. health
  9. spc_latin
```

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins

GCC_Sustainability_Hackathon_R_Comp... | GCC_Sustainability_Hackathon_R_Starte...

Source on Save

```
1  ################################################################################
2  # Title:
3  # Purpose: Download, explore, and analyze a NYC Open Data dataset in R
4  # Author:
5  # Created:
6  # Updated:
7  # Notes:
8  ################################################################################
9  # Step 1 - Prepare a clean R session
10 ################################################################################
11
12
13 if (requireNamespace("rstudioapi", quietly = TRUE) &&
14     rstudioapi::isAvailable()) {
15   rstudioapi::executeCommand("consoleClear")
16 }
17
18 cat("=====================================\n")
19 cat("Step 1: Preparing a clean R session\n")
20 cat("=====================================\n")
21
22 rm(list = ls())
23
24 message("Workspace cleared. Starting clean.")
```

17:1 (Untitled)

Console | Terminal | Background Jobs

R 4.4.3 · ~/scratch_pad/R_code/

```
Rows exported: 1000

CSV file created:
data_out/2015_Street_Tree_Census_selected.csv
# A tibble: 6 × 3
  tree_id zip_city     spc_common
    <dbl> <chr>        <chr>
1  180683 Forest Hills red maple
2  200540 Whitestone   pin oak
3  204026 Brooklyn     honeylocust
4  204337 Brooklyn     honeylocust
5  189565 Brooklyn     American linden
6  190422 New York     honeylocust
>
```

Edit 1: Dataset ID

- In the configuration section, replace uvpi-gqnh with YOUR dataset ID.
- You can leave CATEGORY_COLUMN blank for a first test.
- After viewing the available columns, select one that answers your question.

Edit 2: Category and value

- CATEGORY_COLUMN: a column such as boroname or spc_common.
- VALUE_COLUMN blank: count records per category.
- VALUE_COLUMN numeric: calculate mean value by category.
- Example: category = boroname; value blank → count downloaded tree records by borough.

Run • Inspect • Question

- First run: verify download, row count, and column names.
- Next: inspect the summary for missing, strange, or overly unique categories.
- Finally: view chart.png, and ask whether it actually answers your question.
- Automatic column selection is a starting point, not an analytic decision.

Interpreting data responsibly

- A 5,000-row download may not represent the entire dataset.
- Counting rows may not equal counting people, incidents, or resources.
- Missing values can bias the result.
- Ask: who/what is missing, when was data collected, and what units were counted?

Build your recommendation

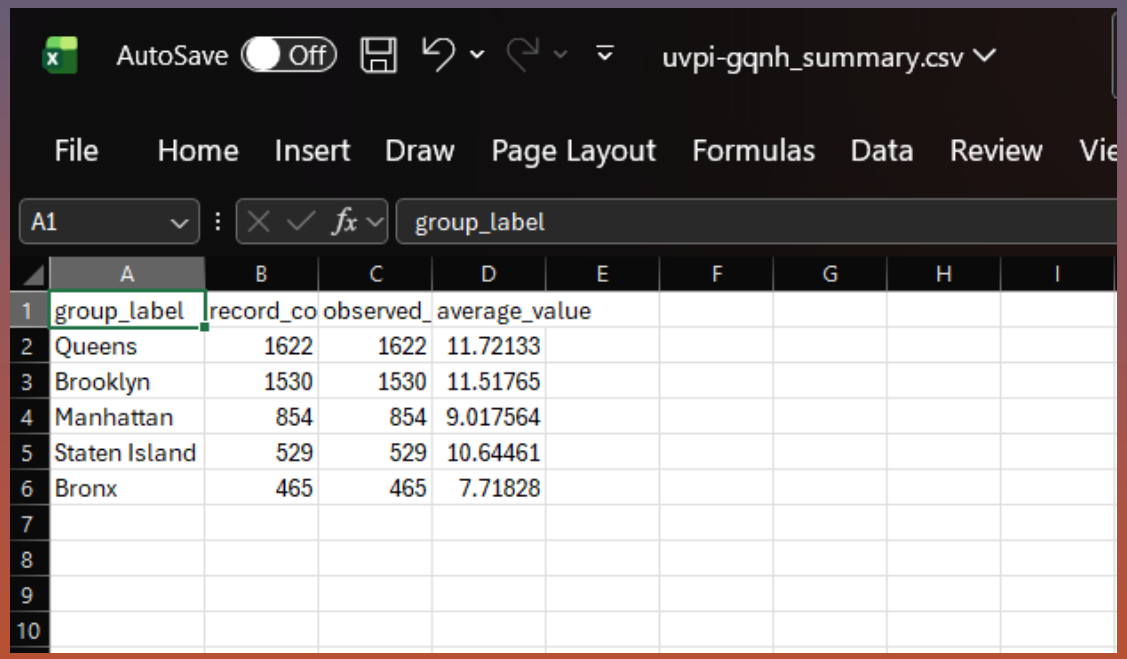
- Observation: What does the chart actually show?
- Interpretation: Why might that pattern matter?
- Limitation: What does the data NOT tell you?
- Action: What practical NYC response do you recommend?
- Next evidence: What else should be studied before implementation?

What to submit

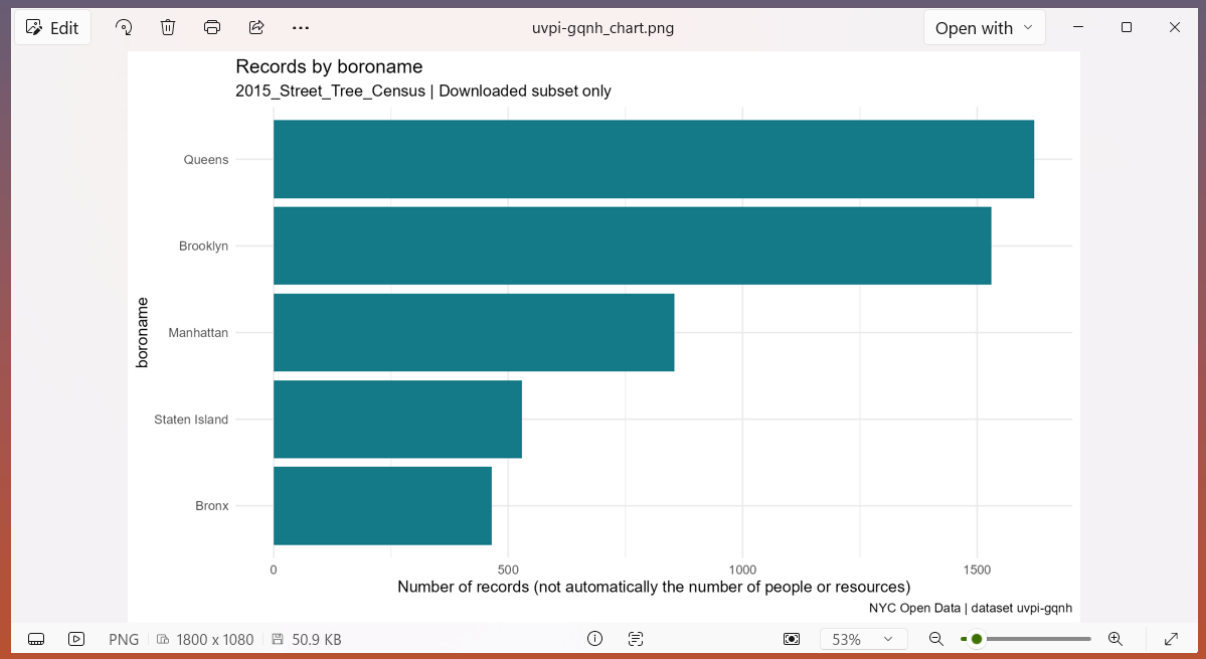
1. Dataset title, URL/ID, and research question
2. R script (.R) OR Python script (.py)
3. One chart (PNG) and summary (CSV)
4. A 3-slide or 2-minute pitch with evidence, limitation, and action

Submissions.

Output: summary.csv and chart.png



	A	B	C	D
1	group_label	record_co	observed_	average_value
2	Queens	1622	1622	11.72133
3	Brooklyn	1530	1530	11.51765
4	Manhattan	854	854	9.017564
5	Staten Island	529	529	10.64461
6	Bronx	465	465	7.71828



How your work is evaluated

- Sustainability problem & relevance — 20%
- Appropriate dataset and responsible interpretation — 25%
- Correct analysis and readable visualization — 25%
- Feasible recommendation and impact — 20%
- Clarity, teamwork, and attribution — 10%

Useful resources

- NYC Open Data: https://opendata.cityofnewyork.us
- NYC API guide: https://www.nyc.gov/opendata/get-started/user-guides/open-data-apis
- RStudio / Posit: https://posit.co/downloads
- Posit Cloud: https://posit.cloud
- Python: https://www.python.org

Hackathon Check List

Student team worksheet

Team name / members: _____

Sustainability challenge: _____

Research question (What, where, or how?): _____

Dataset title / URL / ID: _____

What does each record represent? _____

Chosen category column and optional numeric value:

_____

Main finding (include number and unit): _____

One data limitation (sampling, missingness, geography, date, etc.):

_____

Recommended action and intended beneficiaries:

_____

What would you need to verify before investing in this action?

_____

Final reflection

What did you discover about New York City?

What would you investigate next?

Who could use your findings to make a more sustainable city?

